

Determining Thymol Content in Mouthwash Utilizing Gas Chromatography

Abstract:

Using gas chromatography, a mouthwash sample was analyzed to determine the percent mass of thymol. Standard solutions containing thymol at varying concentrations were analyzed to establish a linear plot. The concentration of the mouthwash sample was calculated using the peak area value generated from the gas chromatograph and the line of best fit. Based on this value and the density of the sample, the percent mass of thymol was computed to be 0.059%, deviating from the labeled value by 0.005%.

Introduction:

Thymol is a compound naturally occurring in the oil of thyme, a culinary and medicinal herb. Shown to possess antioxidant, anti-inflammatory, antibacterial, antifungal, and antiseptic properties; thymol is an active ingredient in disinfectants, medicines, and toiletries, including mouthwashes and toothpastes (1). Thymol can be toxic at high concentrations, so quantitatively analyzing samples is important to ensure product safety and efficacy. Thymol has a relatively low boiling point, allowing it to travel through a gas chromatography column in the gas phase without decomposing.

Gas chromatography is an analytical technique used to separate and quantify components of a mixture—specifically, volatile organic compounds (2). Volatile compounds are those that can be easily vaporized into a gas phase. These compounds are carried through a column with an inert carrier gas (mobile phase), traditionally helium (He), nitrogen (N_2), or Hydrogen (H_2). The gaseous analyte travels through a column coated with a stationary phase, interacting with the column material based on the compound's polarity and volatility. Compounds that interact more strongly with the stationary phase move slowly and elute later. This difference in travel time is

known as retention time. By analyzing a standard, the retention time for a specific compound can be determined and compared with the same compound in an unknown sample for identification.

Each compound in a sample produces a peak when run through a gas chromatograph. This peak is based on the detector's response to the amount of compound present as it elutes from the column. The integral of the response from the detector, or the area under the peak, is referred to as the peak area. This area is proportional to the concentration of the compound present in the sample. A larger peak area indicates a higher concentration, allowing for the generation of a calibration curve given standards under the same conditions.

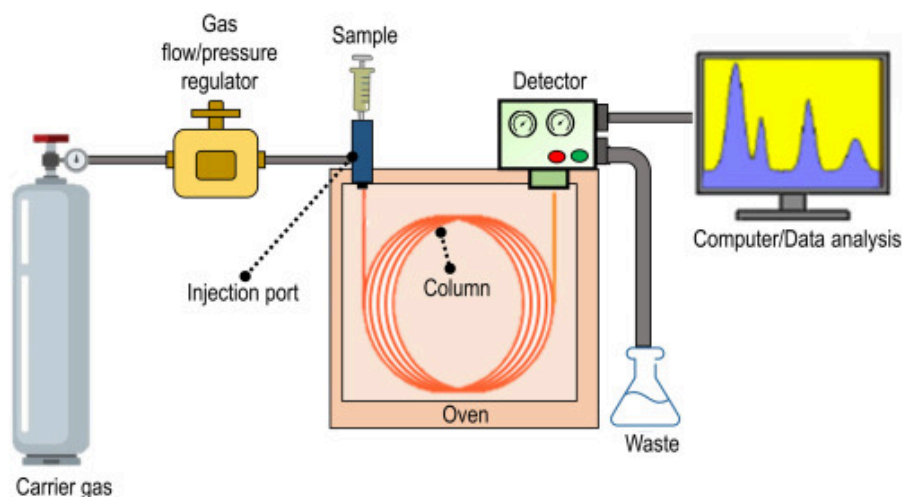


Figure 1: Schematic of a gas chromatograph showing the labeled components (2)

The instrument used in this experiment was the Agilent 6890N Gas Chromatograph (3). The sample is introduced to the instrument using a fine needle, which injects a small volume of the analyte into a pre-heated injection port. The sample is vaporized immediately, then carried through the column with the flow of an inert carrier gas. The oven is a temperature-controlled environment, and as the analyte separates based on interactions with the stationary phase. When the compounds reach the detector is dependent on the retention time, then the detector measures

a signal based on the compound's concentration. These signals are analyzed on the computer to produce a chromatogram, allowing for the identification and quantification of the components in the sample.

Experimental:

A thymol standard was prepared by dissolving approximately 50mg of thymol in 10 mL of methanol, then diluting the solution to 50 mL with additional methanol. Serial dilutions of this stock solution were performed to produce a series of increasingly dilute standards, each with progressively lower concentrations of thymol. Solid-Phase Extraction (SPE) was employed to isolate thymol and similar compounds from a 15 mL mouthwash sample using methanol and water as eluents, bringing the total volume to 30 mL. Gas chromatography, utilizing a packed stainless-steel column, was employed on the standards and sample (from lowest to highest concentration) to establish a calibration curve and thymol retention time. The analyte's peak area was also measured and used to calculate the mass percent of thymol in the mouthwash sample.

Results:

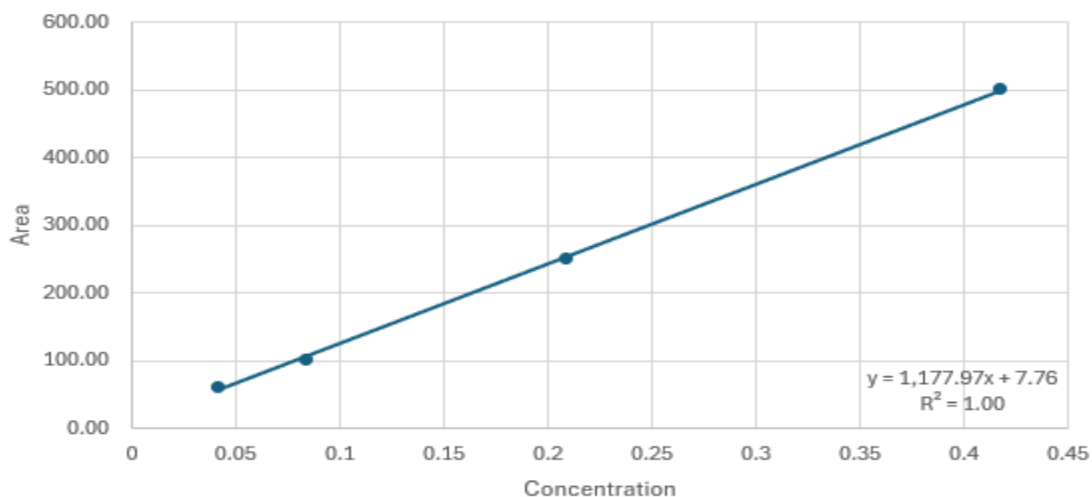
A calibration curve was generated from the peak area and the concentration of the thymol standards. After applying the equation generated from the calibration curve to the peak area of the mouthwash sample, a thymol mass percent of 0.059% was determined.

Initial Amount of Thymol (mg)	52.1
Density of Mouthwash Sample (g/ml)	0.898

Table 1: Initial amount of Thymol for Standards and Density of Mouthwash

Sample	Concentration (mg/mL)	Peak Area
T1	0.0417	62.6
T2	0.0834	101.4
T5	0.2084	250.2
T10	0.4168	500.6
Mouthwash Sample	0.1057	132.3

Table 2: Concentration and Peak Area of Samples and Unknown



Graph 1: Gas Chromatograph Peak Area vs. Concentration of Thymol Standards

To determine the percent mass of thymol present in the mouthwash sample, several calculations were performed. To prepare the stock solution, 52.1mg of thymol was diluted with 50 mL of methanol. This resulted in a concentration of $\frac{52.1\text{mg}}{50.0\text{mL}} = 1.042\text{mg/mL}$ Thymol present in the stock solution. Standards were then prepared from the stock solution, T1, T2, T5, and T10. These solutions used 1, 2, 5, and 10 mL of stock solution, respectively, diluted to 25 mL with methanol. An example calculation for the dilution of 10 mL is as follows:

$$\frac{1.042\text{mg/mL} \cdot 10.0\text{mL}}{25.0\text{mL}} = 0.4168\text{mg/mL}. \text{ After graphing the calculated concentrations vs. the peak}$$

area provided from the chromatographs, the calibration curve $y = 1177.97x + 7.76$ was established. Using the peak area value of 132.6 acquired from the mouthwash sample chromatograph, the concentration (x) was determined to be 0.105 mg/mL. This was then converted to g/L and multiplied by the amount of mouthwash eluted (15 mL), over the total volume of the mouthwash sample (30 mL). After dividing this by the density of the mouthwash solution, the value $\frac{0.106\text{mg/mL}}{0.898\text{mg/mL}} * \frac{15\text{mL}}{30\text{mL}} = 0.059\%$ was produced, which is the mass percent of thymol present in the mouthwash solution.

Discussion and Conclusion:

The experimental results indicate 0.059% mass present in the mouthwash solution, which closely resembles the true value of 0.064%. The deviation of these values is approximately 0.005%, indicating that systematic error likely did not have a sizable role, and random error is the likely source of the deviation. The relative error is 7.81% ppt lower than the true value, further confirming the presence of random error rather than systematic error, which could be mitigated with replicate trials and analyses.

Random error is likely present due to many reasons that make it difficult to pinpoint the specific source. Pipetting inconsistencies are likely the largest factor contributing to the variation from the true value, as there are analyst limitations in reproducing exact volumes at small scales. Environmental temperature and humidity also play a factor, as excess water could react with the hygroscopic methanol (4), diluting the concentration of the samples slowly. Calibration of the gas chromatograph could also lead to minor errors, as with complex instruments, variation in a singular part could lead to a calibration error in the final result.

Gas chromatography is highly accurate and precise, with reliability confirmed through decades of application in countless experiments. The instrument was recalibrated multiple times

during the measurements between samples, cooling the oven to the correct starting temperature to repeat the appropriate sequence for each sample. This increased the accuracy of these measurements, as the environment each sample experienced was consistent. Gas chromatography is also highly sensitive, requiring trace amounts of thymol in a given sample. Since thymol is only approximately 0.064% mass of the mouthwash sample, the instrument's sensitivity is critical for analyzing such a small concentration.

The limitations of gas chromatography are that it is limited to volatile compounds, needs to be calibrated, and can be time-consuming depending on the quantity of samples. Since the analyte needs to be in a gaseous form when run through the column, this can massively limit the types of analytes that can be measured through gas chromatography. Consistent measurements can become problematic if the instrument is not properly calibrated. Different methods have to be established depending on the analyte being measured. Since gas chromatography requires a specific temperature-controlled oven, it is not possible on most instruments to run multiple samples simultaneously, which can become time-consuming when measuring multiple samples.

High-performance liquid chromatography (HPLC), fluorescence spectroscopy, and UV-VIS spectrophotometry are other valid methods that can be employed to detect thymol in a sample. Generally, these methods will still involve the creation of a calibration curve, but dramatically decrease the run-time of multiple samples, leading to less time-consuming and thus less expensive barriers in quantifying the amount of thymol in a sample. The spectral method (UV-VIS, IR, Fluorescence, MS) suggests high sensitivity and a good linear range for analyzing thymol in pharmaceutical preparations (5).

The random error in this experiment could likely be accounted for with multiple trials, allowing for an average and relative standard deviation to be established. Since systematic error

was likely not a major contributing factor in the deviation of the experimental value from the true value, gas chromatography is a validated method for analyzing thymol concentration.

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